

Short Answers1/4
Ans

Computer Vision enables algorithm and AI to always Boost the system and identify objects Patterns and scenes

by objectives

- ⇒ Image understanding
- ⇒ Object detection and recognition
- ⇒ Decision making

by Applications.

- ⇒ Face recognition
- ⇒ Self driving car

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Ans

Image acquisition, PreProcessing, feature extraction Information and Image classification

by Feature extraction: It is the process of identifying important information (edges, corners, textures) from an image.

by Object recognition it identifies and classifies objects, making computer vision subjects easily.

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Ans

by low level Image Processing Basic operations, it noise removal, filtering and image enhanced.

b) Mid level vs High level:

Mid levels: Segmentation & feature extraction

High level: Object recognition and scene understanding.

4) A digital image is represented as a grid of Pixels.

a) Pixel: The smallest unit of a digital image

b) Image resolution: Higher resolution means more pixels and better image quality.

5) A camera sensor: It converts light and converts it into a digital image.

a) Role of Image Sensor:

It converts light into electrical/digital signals.

b) Image acquisition devices

* Digital Camera

* Scanner

6) a) Ans Sampling: Converts a continuous image into discrete pixels.

b) The purpose of quantization in image processing assigns intensity values to each pixel.

7) it explains how light from a scene passes through the camera lenses and it is the process of capturing images.

a) Segmentation & feature extraction that helps take place is called Perspective Projection.

b) Image Processing

* It preprocesses the image, feature extraction information and image, classification

c) Computer Vision.

* Computer Vision is the process of identifying important information.

8) Ans Image acquisition, Preprocessing, feature extraction, information and image classification.

a) feature extraction: It is the process of identifying important information edges, corners textures from an image.

b) object recognition it identifies and classifying objects.

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A digital image represented as grid of Pixels
a) Application in healthcare

The smallest unit of a digital image

b) Application in autonomous vehicles higher
resolution means more Pixels and better image
quality.

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A camera Sensor: It converts light and into digital
image

a) Role of Image Sensor: Converts light into
electrical / digital signals

b) Image acquisition devices.

